



2013

YEAR 11 MATHEMATICS

TERM 2 ASSESSMENT TASK

Date: 7th June, Period 4
Time allowed: 45 minutes (plus 2 minutes reading time)
Total marks: 35 marks

Directions to Candidates

- Attempt all questions.
- Marks are indicated next to each question.
- All necessary working should be shown.
- Board-approved calculators may be used.
- Begin each question on a new page with your student number clearly written at the top.
- Write in black pen and use black pen for all diagrams

Outcomes

A student:

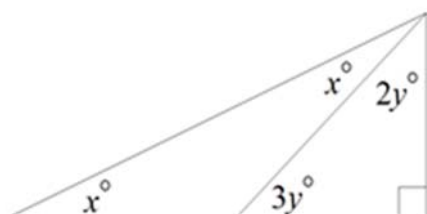
- P2** provides reasoning to support conclusions which are appropriate to the context.
- P3** performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and identities.
- P4** chooses and applies appropriate arithmetic, algebraic, graphical and geometrical techniques.
- P5** understands the concept of a function and the relationship between a function and its graph.

Student Name: _____

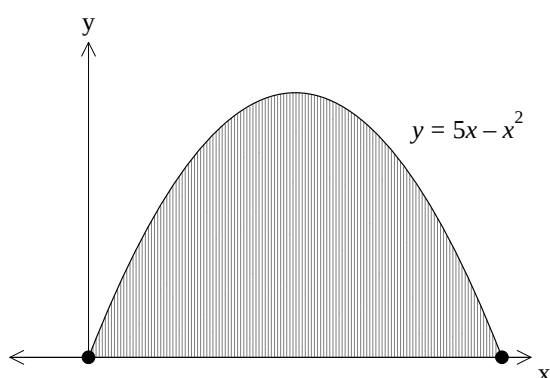
Student Number: _____

Section I
Total Marks (5)
Attempt Questions 1 – 5
Answer Section I on the Multiple Choice Answer Sheet provided.

1. What is the value of x ?



- (A) 18 (B) 27 (C) 36 (D) 45
2. What is the domain and range of the function $f(x) = \sqrt{1-x^2}$?
- (A) Domain: $0 \leq x \leq 1$, Range: $-1 \leq y \leq 1$
 (B) Domain: $-1 \leq x \leq 1$, Range: $-1 \leq y \leq 1$
 (C) Domain: $-1 \leq x \leq 1$, Range: $0 \leq y \leq 1$
 (D) Domain: $0 \leq x \leq 1$, Range: $0 \leq y \leq 1$
3. The diagram shows the graph of the function $y = 5x - x^2$. Which pair of inequalities specifies the shaded region?



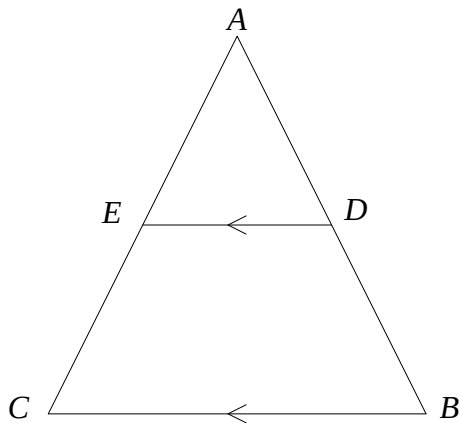
- (A) $y \leq 5x - x^2$ and $y \leq 0$.
 (B) $y \leq 5x - x^2$ and $y \geq 0$.
 (C) $y \geq 5x - x^2$ and $y \leq 0$.
 (D) $y \geq 5x - x^2$ and $y \geq 0$.

4. Consider $f(x) = \frac{6}{x}$ and $g(x) = 2x + 4$.

What are the values for x for which $f(x) = g(x)$?

- (A) $x = -1$ or $x = 3$
- (B) $x = -3$ or $x = -1$
- (C) $x = 1$ or $x = 3$
- (D) $x = -3$ or $x = 1$

5. In the diagram below, ABC is a triangle and DE is parallel to BC .



Given that $AD = 2$, $BD = 5$ and $DE = 1.5$, what is the value of BC ?

- (A) 4.00 (B) 5.25 (C) 7.50 (D) 9.33

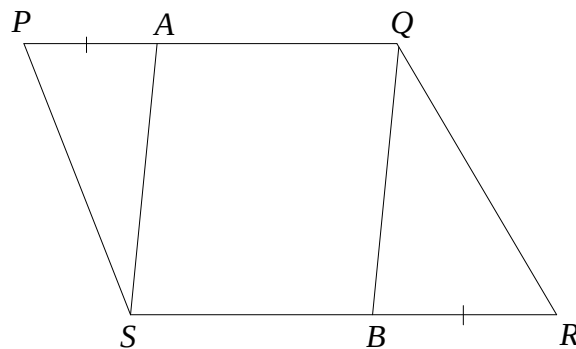
End of Multiple Choice Questions

Section II
Total marks (30)
Attempt questions 6 – 7
All questions are of equal value

Question 6 (15 marks)	START A NEW PAGE	Marks
a) For the function with equation $y = \frac{3}{x+2}$		
i) Write down the equations of the asymptotes.		2
ii) Sketch the graph of this function showing all necessary features.		2
b) i) Find the centre and radius of the circle with the equation $x^2 + 14x + (y-1)^2 - 15 = 0$		3
ii) Does this circle cross the y-axis? Give reasons for your answer.		1
c) Sketch the graph of $y = x-4 + 3$		2
d) The function $f(x)$ is defined as $f(x) = \begin{cases} x^2 + 5, & \text{if } x \geq 0 \\ 5 + x, & \text{if } x < 0 \end{cases}$ Find:		
i) $f(-3) + f(2)$		1
ii) $f(m^2)$		1
e) i) Sketch the graphs of $y = 3^x - 1$ and $y = 3$ on one number plane.		2
ii) Explain why the equation $3^x - 1 = 3$ has only one solution.		1

Question 7 (15 marks) START A NEW PAGE**Marks**

- a) Find the number of sides of a regular polygon with each interior angle equal to 160° ? **2**
- b) The parabola $y = x^2$ is shifted 2 units to the right and 5 units up.
For the shifted parabola:
- i) state the coordinates of the vertex, **1**
 - ii) write down the equation of this parabola, **1**
 - iii) sketch the parabola showing all essential features. **1**
- c) For which values of x is $x^2 + 4x - 21 < 0$? **2**
- d) Show that $f(x) = \frac{x^3}{x^2 - 4}$ is an odd function. **2**
- e) Explain how a function is different from a relation. **2**
Give an example of a function and a relation, where the relation is not a function.
- f) In the diagram below $PQRS$ is parallelogram and $PA = BR$.

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- i) Show that $\triangle PAS \equiv \triangle RBQ$ **2**
- ii) Hence or otherwise show that $AQBS$ is also a parallelogram. **2**

End of paper

Multiple Choice Answer Sheet

Student Number: _____

1 A ☐ B ☐ C ☐ D ☐

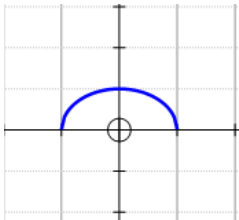
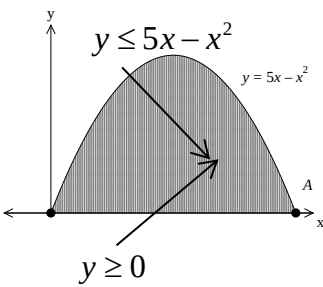
2 A ☐ B ☐ C ☐ D ☐

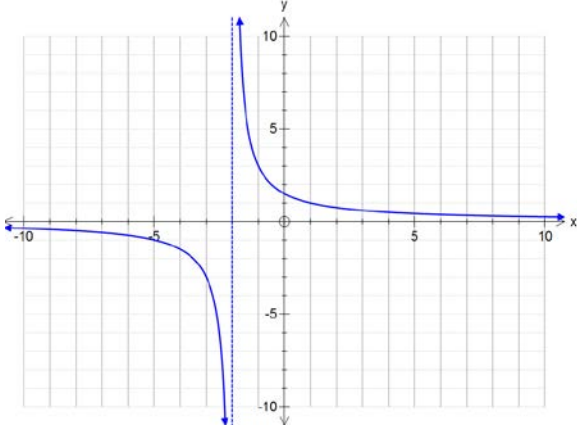
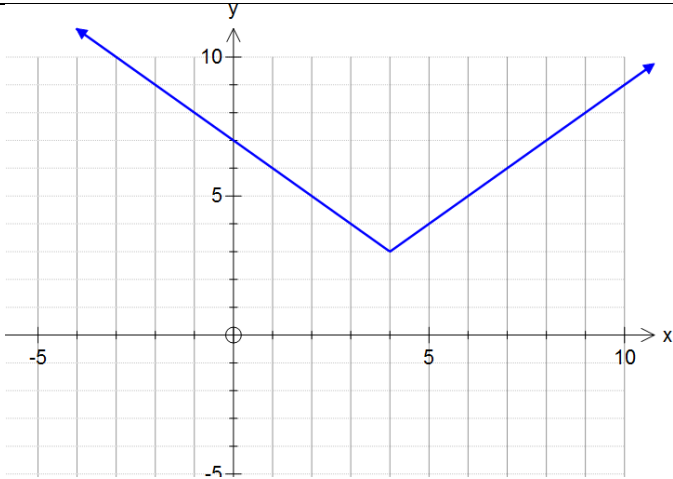
3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

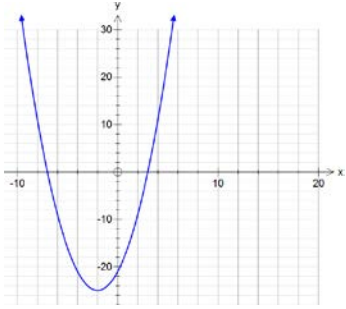
5 A ☐ B ☐ C ☐ D ☐

Year 11 Mathematics Assessment T2 2013
Solutions and Marking Criteria

M/C	Solution	Marking Criteria
1.	$5y = 90$ (angle sum of triangle is 180°) $y = 18$ $2x = 3 \times 18$ (exterior angle of triangle equals sum of 2 opposite interior angles) $2x = 54$ $x = 27$	1 – correct answer B
2.	Upper half of semicircle, with radius one  Domain $-1 \leq x \leq 1$ Range $0 \leq y \leq 1$	1 – correct answer C
3.	 $y \leq 5x - x^2$ $y \geq 0$ $y \leq 5x - x^2$ and $y \geq 0$	1 – correct answer B
4.	$\frac{6}{x} = 2x + 4, x \neq 0$ $2x^2 + 4x - 6 = 0$ $2(x^2 + 2x - 3) = 0$ $2(x + 3)(x - 1) = 0$ $x = -3$ or $x = 1$	1 – correct answer D
5.	$\triangle AED \parallel \triangle ACB$ (equiangular) $\frac{BC}{1.5} = \frac{7}{2}$ (corresponding sides in congruent triangles are in same ratio) $BC = \frac{7}{2} \times 1.5$ $BC = 5.25$	1 – correct answer B

6. a) i)	$x = -2, y = 0$	2 – correct answer 1 – one asymptote correct
a) ii)		2 – correct graph, showing asymptotes and y-intercept 1 – correct graph with asymptotes
b) i)	$x^2 + 14x + (y - 1)^2 - 15 = 0$ $x^2 + 14x + 49 + (y - 1)^2 = 15 + 49$ $(x + 7)^2 + (y - 1)^2 = 64$ Centre $(-7, 1)$, Radius $r = 8$	3 – correct solution 2 – substantially correct solution 1 – correctly completing square or correctly concluding centre or radius from incorrect calculations
b) ii)	Yes, since the centre is 7 units from the y-axis and the radius of the circle is 8 units. Can be shown algebraically by substituting $x = 0$ and showing that $(y - 1)^2 - 15 = 0$ has two solutions, ie y-intercepts.	1- Correct explanation
c)		2 – correct graph 1 – correct shape, not clearly showing intercept or vertex.
d) i)	$f(-3) + f(2) = 5 - 3 + 2^2 + 5$ $= 11$	1 – correct solution
d) ii)	$f(m^2) = m^4 + 5, \text{ since } m^2 \geq 0$	1 – correct answer

e) i)		2 – both graphs correct 1 – one correct graph
e) ii)	Because the two graphs only intersect in one point.	1- correct answer
7. a)	Exterior angle = $180^\circ - 160^\circ$ (supplement of interior angle) $= 20^\circ$ $360^\circ \div 20^\circ = 18$ (exterior angle sum of polygon is 360°) $\therefore$ polygon has 18 sides. OR Angle sum of interior angles in polygon = $(2n - 4)$ right angles $\frac{(2n - 4) \times 90^\circ}{n} = 160$ $180n - 360^\circ = 160n$ $20n = 360$ $n = 18$	2 - correct solution 1 – correctly finding exterior angle OR correctly setting up equation $\frac{(2n - 4) \times 90^\circ}{n} = 160$
b) i)	Vertex = (2, 5)	1 – correct answer
b) ii)	$y = (x - 2)^2 + 5$	1 – correct equation
b) iii)		1 – correct graph, showing y-intercept

c)	$x^2 + 4x - 21 < 0$ $(x - 3)(x + 7) < 0$  $\therefore -7 < x < 3$	2 – correct solution 1 – correct factorisation, OR correct solution from their factors
d)	$f(-x) = \frac{(-x)^3}{(-x)^2 - 4}$ $= \frac{-x^3}{x^2 - 4}$ $= -\left(\frac{x^3}{x^2 - 4}\right)$ $= -f(x)$ $\therefore f(x) \text{ is an odd function}$	2 – correct solution 1 – correct substitution of $(-x)$ and attempt at simplification
e)	In a function every x-value has only one matching y-value, whereas in a relation an x value can have more than one matching y-value. Possible examples: Function: $y = x$ Relation: $x^2 + y^2 = 4$	2 – correct explanation and examples 1 – correct explanation or correct examples for function and relation
f) i)	In $\triangle PAS$ and $\triangle RBQ$ $PA = RB$ (given) $PS = RQ$ (opposite sides in parallelogram are equal) $\angle P = \angle R$ (opposite angles in parallelogram are equal) $\therefore \triangle PAS \equiv \triangle RBQ$ (SAS)	2 – correct proof fully justified 1 – correct proof but not all reasons given or significant attempt at proof
f) ii)	Let $PA = RB = a$ (given) $PQ = RS$ (opposite sides in parallelogram are equal) hence $AQ = PQ - a$ $= RS - a$ $= BS$ $AS = BQ$ (corresponding sides in congruent triangles) $\therefore AQBS$ is a parallelogram (opposite sides are equal)	2 – correct proof fully justified 1 – correct attempt at proof indicated by a correct test for parallelogram OR correct proof not fully justified

Communication			
6b)i)	Clear reasoning based on distance/ graphical or algebraic calculation of y-intercepts	AW 1	
6e)ii)	Clear explanation with correct vocabulary.	AW1	
7b)iii)	Axes labelled and scale accurate	AW 1	
7c)	Shows graph of parabola	AW 1	
7d)	Shows substitution of $(-x)$	AW 1	
7e)	Clear explanation of function	AW 1	